Individual Assignment

Managing and Automating Processes

MSc MDAI — Robotic Process Automation

Academic Year 2025–2026 • Professor: Dr. Isabella SEEBER

Topic 5: Event Ticket Sales and Check-In

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Activity 1: Elaborate on the Organization

1.1 Company Profile

EventPro Solutions is an event management company based in Grenoble, France, specializing in organizing career fairs, professional conferences, and corporate networking events. Founded in 2018, the company manages approximately 40 events per year across the Auvergne-Rhône-Alpes region, serving audiences ranging from 200 to 5,000 attendees per event. EventPro employs 35 full-time staff and hires additional temporary workers for on-site operations during events.

The core business process studied in this assignment is the event registration and check-in lifecycle: from participants registering online through a form, receiving personalized QR-code access passes by email, to on-site QR scanning at venue entrances with time-slot validation.

1.2 IT Strategy

EventPro's IT strategy is built around five key principles:

1. Cloud-based collaboration: All data is stored in Google Sheets and Google Drive, enabling real-time access for all team members regardless of location.

2. Form-driven registration: Google Forms serve as the primary registration interface, eliminating the need for custom web development while collecting structured attendee data.

3. Automated document generation: QR codes and access passes are generated automatically using Google Docs templates and barcode generation services, removing manual document creation.

4. No-code automation (RPA): Make.com (formerly Integromat) orchestrates the entire workflow from registration to check-in, connecting Google Forms, Google Sheets, OneDrive, Google Docs, and email without writing code.

5. Webhook-based check-in: On-site check-in uses webhooks triggered by QR code scanners, enabling real-time validation against the attendee database with time-slot enforcement.

1.3 Strategic Goals

Goal	Target	Link to this assignment
Instant registration processing	Access pass email within 2 minutes of form submission	Automated Scenario 1 (Activity 3)
Reliable check-in	QR scan validation in < 5 seconds, with time-slot enforcement	Automated Scenarios 2 & 3 (Activity 3)
Zero manual document work	100% of access passes auto-generated from templates	QR code + Google Docs template pipeline
Real-time attendee	Live dashboard of check-ins,	Google Sheets updated automatically at every

tracking	entry counts, time stamps	step
Support flexible scheduling	Morning, afternoon, or all-day passes with enforced time windows	Time slot routing in Scenario 2

1.4 Current IT Systems

System	Purpose	Technology
Registration Form	Collect participant data (name, email, organization, time slot)	Google Forms
Attendee Database	Central record of all registrations, QR tokens, check-in status	Google Sheets
QR Code Generator	Create unique entry and exit QR codes per attendee	Make.com Barcode module
Document Service	Generate personalized access pass PDFs from templates	Google Docs + Templates
Cloud Storage	Store generated QR code images for embedding in documents	OneDrive
Email Service	Deliver access pass PDFs to participants	Microsoft Outlook (via Make.com)
Check-In System	Validate QR scans via webhooks with real-time database lookup	Make.com Webhooks + Google Sheets
Automation Platform	Orchestrate all workflows across systems	Make.com

1.5 Pain Points

The following pain points exist in the current manual process and motivate the automation described in Activity 3:

Pain Point 1 — Manual access pass creation: Without automation, staff must manually generate a QR code for each registrant, insert it into a document template, convert to PDF, and email it. For a 500-attendee event, this takes approximately 6–8 hours of repetitive work and is highly error-prone.

Pain Point 2 — No time-slot enforcement at check-in: Participants who registered for a morning session could arrive in the afternoon (or vice versa) without any system to detect this. Staff had to manually check spreadsheets, leading to long queues and human errors.

Pain Point 3 — No real-time check-in tracking: Event managers had no live visibility into how many participants had actually arrived. Entry counts were compiled manually at the end of the day, making it impossible to react to no-shows or overcrowding in real time.

Pain Point 4 — Re-entry management: Participants who left the venue temporarily (e.g., for lunch) had no mechanism to re-enter without going through the full check-in process again. This caused confusion and duplicate entries in the attendance records.

Pain Point 5 — Data silos between registration and check-in: Registration data sat in one spreadsheet, check-in data in another. Reconciling them after the event required hours of manual work and produced inconsistencies.

Activity 2: Model a Business Process with BPMN

2.1 Process Overview

The BPMN model covers the full event registration and check-in lifecycle, divided into two phases:

- **Phase 1 — Online Registration:** A participant fills out the registration form and selects a time slot (morning, afternoon, or all day). The system adds them to the database, determines valid hours, generates unique entry and exit QR codes, creates a personalized access pass document, and sends it by email.
- **Phase 2 — On-Site Check-In:** On event day, the participant shows their QR code at the entrance. The system scans it via a webhook, searches the attendee database, validates the time slot, and either grants or denies entry — updating the check-in status and entry count in real time.

2.2 Model Structure

Element	Details
Pools (3)	Participant | Event Management System | Cloud Services (OneDrive, Google Docs, Outlook)
Lanes (2, in Event System)	Registration Platform (Automated) | On-Site Check-In System (Automated)
Tasks (21)	5 in Participant + 10 in Registration + 6 in Check-In + 3 in Cloud (see step-by-step below)
Exclusive Gateways (6)	Registration confirmed? | Registration valid? | Which time slot? (3 branches) | (merge) | Registration found? | Within valid hours?
Message Flows (6)	Registration data | Rejection notice | QR + pass data | Access pass email | QR code scan | Entry granted
Data Objects (4)	Registration Form Data | QR Codes (Entry + Exit) | Access Pass (PDF) | Attendee Record
Data Store (1)	Google Sheets Database
Start Events (3)	1 plain (Participant) + 2 message (Event System, Cloud Services)
End Events (7)	Registration rejected | Attending event | Rejected | Registration complete | Entry granted | Entry denied | Pass delivered

2.3 Step-by-Step Process Flow

Participant Pool:

Step	Task	Then

1	Fill registration form (Google Form)	→ Receive email response
2	Receive email response	→ Gateway: Registration confirmed?
3a	If No: End (Registration rejected)	
3b	If Yes: Present QR code at entrance	→ Enter event → End (Attending event)

Event Management System — Registration Platform Lane:

Step	Task	Then
1	Form submission received (message start)	→ Add registration to Google Sheet
2	Add registration to Google Sheet	→ Gateway: Registration valid?
3a	If No: Send rejection email → End	
3b	If Yes: Determine time slot	→ Gateway: Which time slot?
4a	Morning: Set valid hours (08:00–12:00)	→ merge
4b	Afternoon: Set valid hours (13:00–17:00)	→ merge
4c	All Day: Set valid hours (08:00–17:00)	→ merge
5	Generate entry and exit QR codes	→ Update registration status → End (Registration complete)

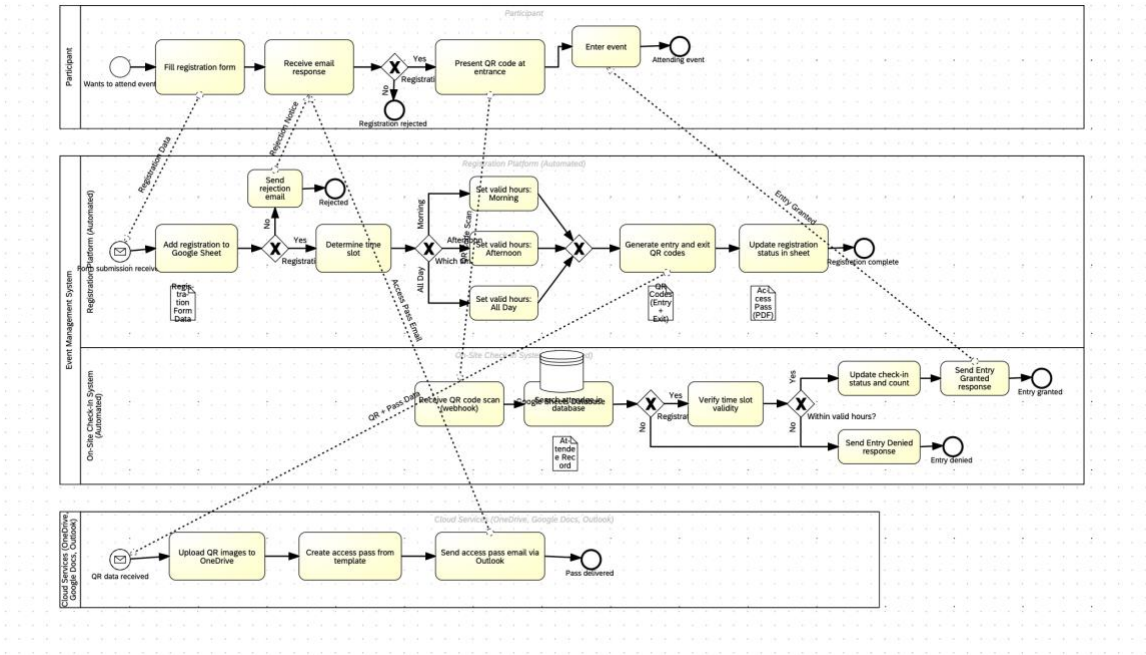
Event Management System — On-Site Check-In Lane:

Step	Task	Then
1	Receive QR code scan (webhook)	→ Search attendee in database
2	Search attendee in database	→ Gateway: Registration found?
3a	If No: Send Entry Denied response → End	
3b	If Yes: Verify time slot validity	→ Gateway: Within valid hours?
4a	If No: Send Entry Denied response → End	
4b	If Yes: Update check-in status and count	→ Send Entry Granted response → End

Cloud Services Pool (OneDrive, Google Docs, Outlook):

Step	Task	Then
1	QR data received (message start)	→ Upload QR images to OneDrive
2	Upload QR images to OneDrive	→ Create access pass from template
3	Create access pass from Google Docs template	→ Send access pass email via Outlook → End

2.4 BPMN Diagram



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2.5 Signavio Link

https://academic.signavio.com/p/editor?id=64ac6399291943f9862511465f6a8e44

2.6 Rationale for Modelling Decisions

Decision 1: Time-slot routing with 3-branch gateway

The “Which time slot?” exclusive gateway splits the flow into three branches (Morning, Afternoon, All Day), each setting different valid hours before merging back. This directly reflects Scenario 2 in the Make.com implementation, which uses a Router with three filtered paths to update the Google Sheet with the correct "valid from" and "valid to" values. The alternative would be a single task with an internal conditional, but the 3-branch gateway makes the business logic visible and easier for the teacher to trace to the actual automation.

Decision 2: Separate Cloud Services pool

OneDrive, Google Docs, and Outlook are modeled as a separate “Cloud Services” pool rather than as tasks within the Event Management System. This was done because these are external third-party services that EventPro does not control they have their own processing logic (file storage, template rendering, email delivery). Modeling them as a separate pool accurately represents the message-based integration between EventPro’s automation and these external APIs, which is visible in the Make.com scenario as separate modules (OneDrive: Upload, Google Docs: Create from Template, Outlook: Send Email).

Decision 3: Excluded — Exit/re-entry process

The full system includes an exit QR scanning process (Scenario 3 in Make.com) that resets the entry flag so participants can re-enter the venue. This was excluded from the BPMN model because: (a) it is a separate sub-process triggered by a different QR code (exit vs. entry); (b) including it would add a second check-in sub-flow with its own webhook, database search, and update logic, making the diagram significantly more complex without demonstrating new BPMN constructs; (c) the core business logic (registration → QR generation → check-in validation) is fully represented without it.

Activity 3: Automate a Part of the Process Using RPA

3.1 Pain Points Addressed

The automation addresses Pain Points #1 (manual access pass creation), #2 (no time-slot enforcement), and #3 (no real-time tracking). Three Make.com scenarios work together to automate the entire registration-to-check-in pipeline:

3.2 Automation Platform

Make.com (formerly Integromat) was used instead of Power Automate because: (a) it provides the same RPA capabilities — visual scenario builder, conditional routing, multi-system connectors; (b) the Webhook module in Make.com is essential for the QR code scanning integration, which would require a premium Power Automate connector; (c) Make.com is freely accessible for students without a Microsoft 365 enterprise license. The core automation patterns (trigger → condition → action) are identical across both platforms.

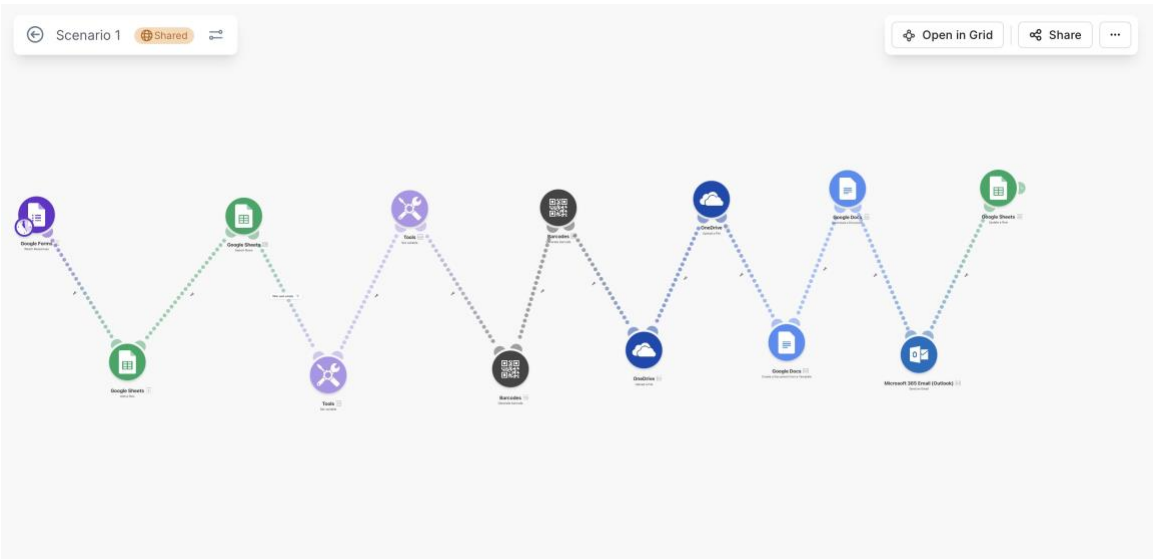
3.3 Scenario 1: Registration and Access Pass Generation

This scenario automates the entire registration flow — from form submission to email delivery of a personalized access pass with QR codes.

#	Module	System	What it does
1	Google Forms: Watch Responses	Google Forms	Triggers when a participant submits the registration form.
2	Google Sheets: Add Row	Google Sheets	Adds the participant's data (name, email, organization, time slot) to the Attendee Database sheet.
3	Google Sheets: Filter Rows	Google Sheets	Retrieves the newly added row to get the row number and all fields for later modules.
4	Set Variable: qr_url	Make.com	Builds the unique entry QR webhook URL using the participant's unique token (column 8).
5	Set Variable: qr_out	Make.com	Builds the unique exit QR webhook URL for the same participant.
6	Barcode: Generate (entry)	Make.com	Generates an entry QR code image from the webhook URL.
7	Barcode: Generate (exit)	Make.com	Generates an exit QR code image.
8	OneDrive: Upload (entry)	OneDrive	Uploads the entry QR image to OneDrive cloud storage.
9	OneDrive: Upload (exit)	OneDrive	Uploads the exit QR image.

10	Google Docs: Create from Template	Google Docs	Creates a personalized access pass document by inserting the participant's name, ID, and both QR images into a pre-designed template.
11	Google Docs: Export	Google Docs	Exports the access pass as a PDF.
12	Outlook: Send Email	Outlook	Sends the PDF access pass to the participant's email with the subject "Access Pass".
13	Google Sheets: Update Row	Google Sheets	Updates the database: stores the QR URL, marks "email sent = Yes", resets entry flag to 0.

Filter (condition): The Set Variable module has a filter called "filter sent emails" that checks whether the email has already been sent for this registration, preventing duplicate emails if the scenario runs again.



After creating a google form and a landing page we built this scenario to be able to send the QR_codes pdf (one for entry and one for exit) and update the google sheet as well.

Scenario 1.blueprint.json

3.4 Scenario 2: Time Slot Update

This scenario runs after Scenario 1 and updates the "valid from" and "valid to" columns based on the participant's chosen time slot.

#	Module	System	What it does
1	Google Sheets: Watch New Rows	Google Sheets	Triggers on new rows where email has been sent.
2	Router (3 paths)	Make.com	Routes based on the time slot field:
3a	Google Sheets: Update Row (Morning)	Google Sheets	Sets valid_from = 08:00, valid_to = 12:00.
3b	Google Sheets: Update Row (Afternoon)	Google Sheets	Sets valid_from = 13:00, valid_to = 17:00.

3c	Google Sheets: Update Row (All Day)	Google Sheets	Sets valid_from = 08:00, valid_to = 17:00.

Condition: The Router uses three filtered paths — each filter checks the value of the time slot column (Morning / Afternoon / All Day) and routes to the corresponding Update Row module.

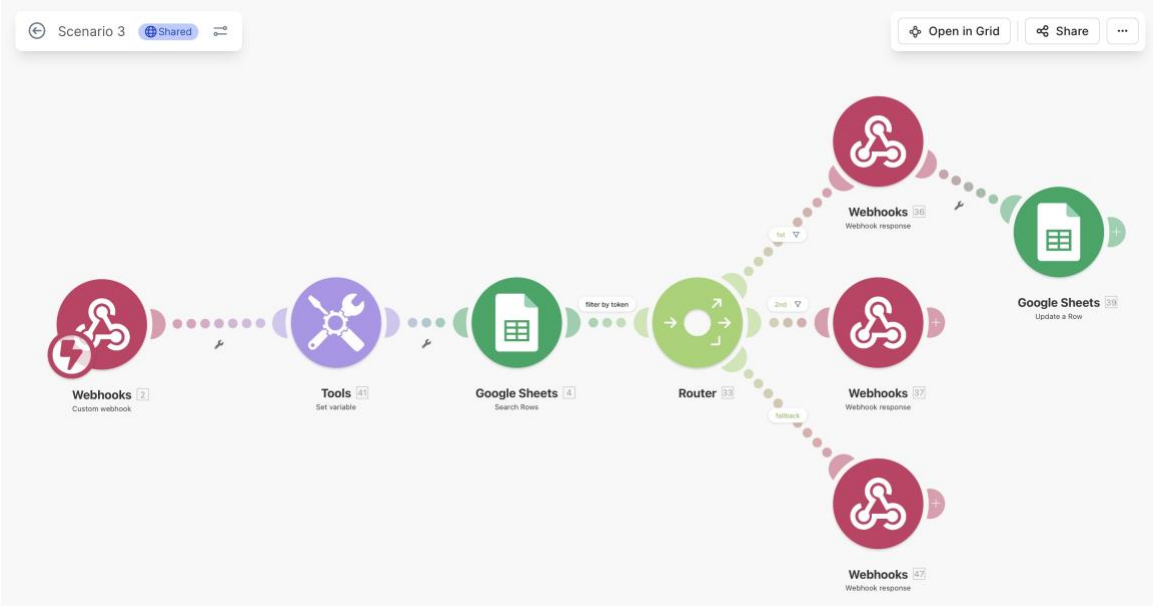
3.5 Scenario 3: Entry Check-In via QR Scan

This scenario handles the on-site check-in when a participant scans their entry QR code at the venue entrance.

#	Module	System	What it does
1	Webhooks: Custom Webhook	Make.com	Listens for incoming QR scan requests. The QR code encodes a URL that triggers this webhook with a unique token parameter.
2	Set Variable: token	Make.com	Extracts the token from the webhook request.
3	Google Sheets: Filter Rows	Google Sheets	Searches the Attendee Database for a row where the QR token matches.
4	Router (2 paths)	Make.com	Routes based on validation conditions.
5a	Webhooks: Respond “Entry Granted”	Make.com	(Valid path) Returns a JSON response with status ACCEPTED, participant name, time slot, and entry count. Then updates the sheet.
5b	Google Sheets: Update Row	Google Sheets	(Valid path) Marks the participant as checked in, increments the entry count, records the check-in timestamp.
6	Webhooks: Respond	Make.com	(Invalid path) Returns a JSON response with status

	"Entry Denied"		REJECTED and the reason.

Conditions: The valid path checks three things simultaneously: (a) the entry flag = 0 (participant is not already inside), (b) the current time ≥ valid_from, and (c) the current time ≤ valid_to. If any condition fails, the denied path sends the rejection response.



3.6 Two Systems + Data Exchange + Conditions

Each scenario exchanges data between at least two systems and includes at least one condition:

Scenario	Systems	Condition	Data Exchanged
Scenario 1	Google Forms → Google Sheets → OneDrive → Google Docs → Outlook	Filter: email not already sent	Registration data, QR images, PDF pass, email
Scenario 2	Google Sheets (read) → Google Sheets (update)	Router: Morning / Afternoon / All Day	Time slot → valid_from and valid_to values
Scenario 3	Webhook → Google Sheets → Webhook response	Router: token match + time validation + entry flag	QR token, attendee record, check-in timestamp

3.7 Workarounds and Assumptions

What	Workaround	Why
Event website	Google Form replaces it	Collects same fields: name, email, organization, time slot
Attendee database	Google Sheet replaces a SQL database	Sheet columns act as database fields; Filter Rows acts as SQL queries
Ticketing/payment	Free registration (no payment step)	Career fairs are typically free; focus is on check-in management
QR code scanner	Webhook URL encoded	Scanning the QR code opens the URL, triggering the

	in QR	Make.com webhook automatically
Power Automate	Make.com replaces it	Same trigger → condition → action pattern; webhook support needed for QR scanning
Venue door system	Webhook response	The JSON response (ACCEPTED/REJECTED) could drive a physical gate in production

Activity 4: Screencast Recording

A ~3 minute screencast demonstrates the full automation pipeline:

Segment	Duration	Content
1. BPMN walkthrough	3 mn	Open Signavio. Walk through the 3 pools: Participant registration, Event System (registration lane + check-in lane), and Cloud Services. Highlight gateways, message flows, and the time-slot routing.
2. Run Scenario 1,2	3mn	Open Make.com → Run Scenario 1 → submit a test Google Form entry → show green checkmarks on all 13 modules → open email inbox → show received access pass PDF with QR codes.
3. Run Scenario 3	3mn	Simulate a QR scan by opening the webhook URL in a browser → show "Entry Granted" JSON response → open Google Sheet → show updated check-in status, entry count, and timestamp.

Video link

https://drive.google.com/file/d/1fAfVH2a1tYQQr4shBWZbpq84_GpjDl03/view?usp=share_link

Appendix: Setup References

A. Google Form Fields

Field	Type	Options/Validation	Required
Full Name	Short answer	—	Yes
Surname	Short answer	—	Yes
Email Address	Short answer	Email validation	Yes
Time Slot	Multiple choice	Morning / Afternoon / All Day	Yes
Organization	Short answer	—	No

B. Google Sheet Columns (Attendee Database)

Column	Content	Updated by
A	Full Name	Scenario 1 (from form)
B	Surname	Scenario 1 (from form)
C	Email	Scenario 1 (from form)
D	Time Slot	Scenario 1 (from form)
E	Organization	Scenario 1 (from form)
F–G	Unique ID / Token	Scenario 1 (auto-generated)
H	Entry QR URL	Scenario 1
I	Exit QR URL	Scenario 1

J	Email Sent (Yes/No)	Scenario 1
K	Entry Count	Scenario 3 (incremented on each check-in)
L	Last Check-In Timestamp	Scenario 3
M	Check-In Status	Scenario 3 (Checked In / —)
N	Entry Flag (0/1)	Scenario 3 (1 = inside, 0 = outside)
O–P	Valid From / Valid To	Scenario 2

C. Signavio Import Instructions

1. Log into academic.signavio.com → Create a new folder. 2. Try Import → Upload the EventPro_Process.bpmn file. 3. If import works, use Format → Auto-Layout to clean up positions. 4. If import fails, rebuild manually following Section 2.3 above. 5. Export as PNG → Paste into Section 2.4. 6. Share → Copy Link → Paste into Section 2.5.